Condition Assessment of Structural Materials using Non-Destructive Examination and Machine Learning

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Abstract

Aging infrastructure, particularly timber piles and similar structural elements, face increased degradation due to factors like rot, decay, and various environmental stressors on top of regular wear. Crucially, significant damage often occurs internally or in inaccessible areas, making it undetectable by conventional visual inspections. This hidden deterioration can lead to unpredictable structural failures, escalated repair costs, and compromised safety across various sectors. This research aims to address this challenge by analyzing acoustic-vibration signals and developing a machine learning-driven system for continuous health evaluation of infrastructure, specifically timber piles. This work provides a vital tool for proactive maintenance strategies, enabling earlier detection of deterioration and ultimately extending the service life of critical infrastructure assets.

Methods

- **A total of 14,829 acoustic recordings were collected from 26 timber piles under the following procedures:**
  - Piles were marked and tapped at 12-inch intervals. In areas near exposed bolts, markings were made every 3 inches for ± 1' from the bolt.
  - Tapping was performed along each orientation of every pile (0°,90°,180°, 270°).
  - Three different tapping devices were used at each location: a claw hammer, a helicopter hammer, and a rubber mallet.
  - This process was repeated three times for each hammer, resulting in 9 audio recordings per location.
- Ground Truth
  - The ground truth health label ('Good', 'Medium', 'Bad') for each pile was determined through **visual inspection** (Figure 2).
- Model Training
  - Acoustic Data recorded **within 30 inches of the visually inspected area** was used to train a 1D Convolutional Neural Network to predict pile health at any location along the pile (Figure 3).

Highlighted area designating where we used visual inspection as ground truth for our data.

Figure 1: Photo of one of the Timber Piles (#6).

Figure 2-A: Good Health Pile. Solid wood with minimal cracking or signs of decay.

Figure 2-B: Medium Health Pile. Cracks or other noticeable signs of stress or damage is visible.

Figure 2-C: Bad Health Pile. Widespread internal rot, often visible as large open gaps.

Figure 2: Visual ground-truth classifications based on pile cross-sections.

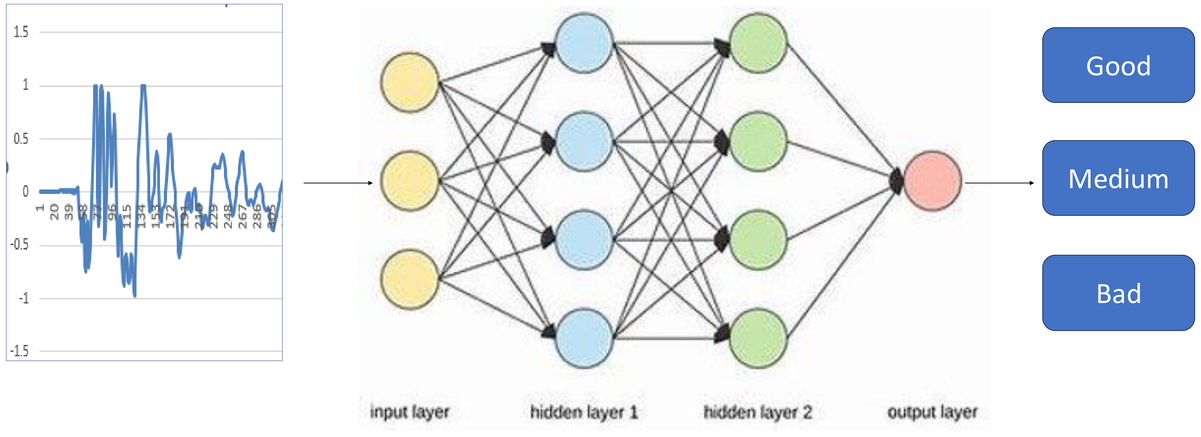

Figure 3: Feature Extraction from Filtered Audio Data into Convolutional Neural Network (CNN).

Outcomes

- Initial methods were insufficient. An early approach using hand-engineered acoustic features and unsupervised clustering failed to reliably distinguish between the different pile health conditions.
- A 1D **Convolutional Neural Network** (CNN), trained directly on the raw vibration signals, rather than the signals, achieved a 66% baseline accuracy, confirming that a learnable pattern exists in the data.
- These results establish that the deep-learning approach is currently the most promising direction, as the acoustic emissions from the piles are **too complex for simpler, feature-based models.**

	Precision	Recall	F1-Score	Support
Good	0.67	0.59	0.63	95
Medium	0.68	0.69	0.69	126
Bad	0.65	0.69	0.67	131

Table 1

Conclusion/Future Works

- The 66% accuracy achieved by the CNN serves as a strong proof-of-concept, demonstrating that acoustic analysis has potential to be a viable, cost-effective method for non-destructive health monitoring.
- Another aspect of improving our health predictions involves incorporating a more **objective ground truth** such as density or compression testing to improve model accuracy and robustness without increasing complexity.
- We are also developing a **multi-modal system** integrating infrared thermal image data (Figure 4) to provide additional information for our prediction model.

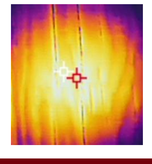

Figure 4: Captured Infrared Thermal Image of Pile.

Acknowledgments

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